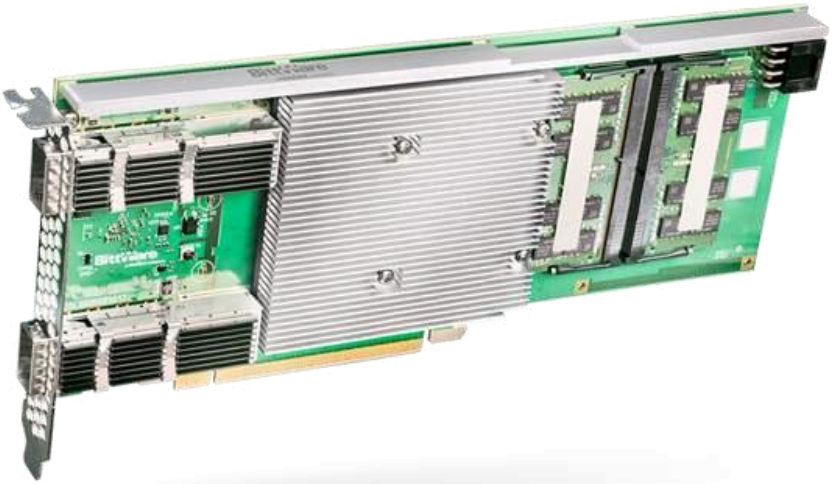

BMC 3.0

Board Management Controller

Reference Guide

BittWare

a molex company

BMC Reference Guide

Release Date: 11/11/24

BittWare
45 South Main Street, Suite L100
Concord, NH 03301 USA
603.226.0404

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1 Introduction

1.1 About this Reference Guide

1.1.1 Purpose of this Document

This reference guide provides the details needed for low-level communications with the Board Management Controller (BMC). The BittWare SDK and individual Card Support Packages (CSP) contains software utilities and libraries for communicating with the BMC at a higher, more abstract level.

1.1.2 Document History

Table 1: Document Changes and Updates

Date	Description of Changes
14-Nov-2023	Initial Revision
25-Jan-2024	Added IA-860m sensors Added some missing PLDM OEM Message descriptions
11-Nov-2024	Added IA-720i

1.2 Release Notes

BMC 3.0 release notes are available to download on the BittWare developer site.

1.3 Related Documentation

Refer to the following documents (available on the BittWare developer site) for more information about the card specific BMC details

- Intel Agilex FPGA documentation

1.4 Abbreviations & Definitions

- BMC Board Management Controller
- CSP Card Support Package
- CvP Configuration Via Protocol
- DDR Double Data Rate
- FPGA Field Programmable Gate Array
- Gb Gigabit
- GB Gigabyte
- I2C Inter-IC bus (standard)
- IO Input/Outputs
- IP Intellectual Property
- JTAG Joint Test Action Group (boundary scan standard)
- LED Light Emitting Diode
- MCTP Management Component Transfer Protocol

- MT/s Mega Transfers per second
- PLDM Platform Level Data Model
- POF Programmer Object File
- PR Partial Reconfiguration
- QSFP Quad Small Form-factor Pluggable
- QSFP28 28GHz version of QSFP
- SDRAM Synchronous Dynamic Random-Access Memory
- SOF SRAM Object File
- SPI Serial Peripheral Interface
- PROM Programmable Read-Only Memory
- QSPI Quad Serial Peripheral Interface
- TCXO Temperature Compensated X-tal (crystal) Oscillator
- VID Voltage IDentification

1.5 Contacting BittWare

BittWare is dedicated to providing customers with superior technical support:

- **BittWare Developer Site:** The BittWare Developer Site provides online access to our technical support resources. Go to developer.bittware.com to register for an account. Once you have set up an account, you will have access to BITTS (the BittWare Issue Tracking and Technical Support site), BittWare product documentation, software downloads, release notes, and examples. When you are logged into the Developer Site, follow the "Issue Tracking" link at the top right of the screen to access BITTS.
- **BittWare's website:** Our website at www.bittware.com provides a variety of literature, including whitepapers, datasheets, and articles.
- **Phone**: You can also call us directly at +1 (603) 226 0404 between the hours of 8:30 a.m.– 5:00 p.m. (US Eastern Time) or +44 (0) 1236 373500 between the hours of 8:30 a.m. – 5:00 p.m. (UK GMT).

1.6 Customer Feedback

Thank you for using BittWare products. We appreciate you choosing BittWare for your FPGA development.

If you have a few minutes to spare, we would love to hear from you about your experience with our products or our staff. We know your time is valuable; we would be grateful for any comment at all.

- Please email us at support@bittware.com for any type of feedback
- Or, let us know about a recent Technical Support inquiry at http://koch.link/bwsupportfeedback

2 BMC Overview

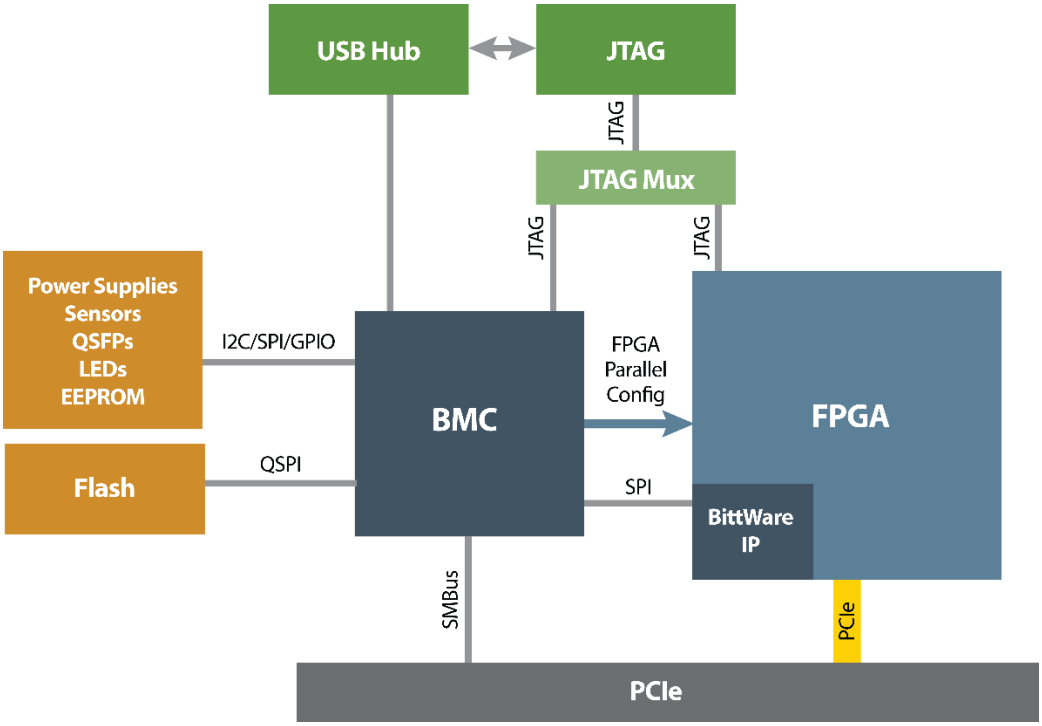
The Board Management Controller (BMC) is responsible for controlling, monitoring, and giving access to low level functions of a BittWare device and its peripherals. On BittWare cards, the BMC is implemented with a microcontroller. The device provides a unique set of functions and abilities. The microcontroller interfaces with on-board sensors, the optional DIMMs, the FPGA, and the Flash, and it controls power and resets. The microcontroller communicates using the Management Component Transport Protocol (MCTP) with payloads defined by the Platform Level Data Model (PLDM) specification. The BMC uses some MCTP and PLDM vendor specific commands for BittWare-specific extensions.

The BMC 3.0 includes card-level security features such as key management and signed firmware images. Security is customer-configurable, which allows you to implement the level of security you need for your application.

The BMC functions as a Root of Trust and can be extended to provide security features around the FPGA and flash. The FPGA bitstream storage in the flash is behind the BMC, accessible via BittWare SDK utilities. All I^2C peripherals are accessible only through the BMC instead of directly from the FPGA to the I^2C bus. The BMC can also be configured to disable external JTAG access.

This reference guide provides the details needed for communicating with the BMC at a low level. The BittWare SDK and Card Support Packages (CSP) contains utilities and libraries for communicating with the BMC at a higher, more abstract level.

Figure 1: BMC Interface Overview



2.1 Microcontroller Summary

The microcontroller (MCU) interface of the Board Management Controller (BMC) is responsible for controlling, monitoring, and giving access to low level functions of a BittWare device and its peripherals. The microcontroller implements the MCTP/PLDM protocol and BittWare's custom command extensions. MCTP/PLDM commands can be issued to the BMC over multiple interfaces.

2.1.1 Microcontroller Features

The microcontroller provides the following:

- Power sequencing and control for the board
- Voltage, current, temperature monitoring
- Ability to configure programmable clocks and change the default programs
- Presence detection and access to other peripheral components, usually via I^2C
- Field upgrades of BMC firmware
- Programming/Configuration of the FPGA from Flash

2.1.2 How to Access the Microcontroller Interface

There are three ways to access the microcontroller interface of the BMC:

1. **USB**: The card USB connector is attached to the BMC microcontroller. MCTP/PLDM packets are used over this interface to communicate with the BMC.
2. **FPGA PCIe**: The BMC is connected to the FPGA. BittWare provides FPGA IP that will provide a communication path with the BMC over PCIe. See the card's FPGA Developer Guide for further details.
3. **SMBus**: An SMBus connects the server's BMC with the BMC on the card. When this connection is enabled, MCTP/PLDM over SMBus protocol is used.

2.1.3 BMC Software Libraries

The primary software for interfacing with the BMC is 3 Python packages:

- bw-mctp (MCTP protocol)
- bw-pldm (PLDM protocol)
- bw-bmc (abstraction layer over the PLDM protocol)
  The SDK provides a "C" binding for bw-bmc with a subset of the features provided in the bw-bmc python package. (see SDK libbwsdk_bmc)

There are also 3 BittWare SDK Python packages that provide the physical interface for communicating with the BMC:

- bw-usb (USB communication)
- bw-pci (PCI communication used in conjunction with bw-agilex)
- bw-agilex (Agilex FPGA family library package)

3 SMBus Commands

3.1 SMBus Specification Compliance

The BMC conforms to the SMBus Specification Version 2.0.

3.2 SMBus Commands

The BMC responds on the SMBus as an *ARP-capable* device. Its default I^2C slave address is 0xCE. The BMC will respond with a Unique Device Identifier (UDID) with fields set according to the table below:

Table 2: Unique Device Identifier fields

Field name	Description	Value
Device Capabilities	Dynamic and Persistent, PEC supported	0x41
Vendor / Revision	UDID Version 1, Silicon Revision ID 1	0x09
Vendor ID	BittWare PCI Vendor ID	0x12ba
Device ID	Card PCI Device ID	IA-860m: 0x76 IA-780i: 0x75 IA-440i: 0x73 IA-720i: 0x78
Interface	ASF, SMBus 2.0	0x24
Subsystem Vendor ID	BittWare PCI Vendor ID	0x12ba
Subsystem Device ID	Card PCI Device ID	IA-860m: 0x0076 IA-780i: 0x0075 IA-440i: 0x0073 IA-720i: 0x0078
Vendor-specific ID	Serial number	Unique per board

The BMC implements the following SMBus commands:

- Prepare to ARP
- Reset device (general)
- Get UDID (general)
- Assign address
- Get UDID (directed)
- Reset device ARP (directed)

3.3 SMBus MCTP Binding

The BMC implements the MCTP SMBus/I2C Transport Binding on the PCIe/SMBus interface. See DTMF DSP0237 for details.

3.4 Enable SMBus Interface

The SMBus interface is disabled by default to prevent interference on systems that don't support ARP and possibly have an existing I2C/SMBus device that is using the BMC default address. To enable the SMBus interface use the **bw_bmc_configure** command to enable the interface. This setting is persistent and only needs to be done once. This same command can be used to disable the interface.

3.4.1 Show current interface setting

```
bw_bmc_configure pci
PCIe SMBus:   Disabled
```

3.4.2 Toggle interface enable/disable

```
bw_bmc_configure pci -i
PCIe SMBus:   Enabled
```

4 MCTP

All communication with the BMC uses MCTP. The BMC does not conform to the MCTP specification in the following ways:

- MCTP messages are not limited to the baseline transmission unit size of 64 bytes

4.1 MCTP Control Messages

The BMC implements the following MCTP Control messages,

- Set Endpoint ID
- Get Endpoint ID
- Get Endpoint UUID
- Get MCTP Version Support
- Get Message Type Support
- Get Vendor Defined Message Support

4.2 MCTP Vendor Defined Messages

The MCTP Vendor Defined messages use the Vendor Defined – PCI message format. These commands are not intended to be implemented directly by users, so details are not published. The BMC implements the following MCTP Vendor Defined messages:

- Echo

5 PLDM

PLDM is used for communication of sensor data and other command and control of the BMC.

5.1 PLDM Base Specification Commands

The BMC implements the following PLDM Base Specification commands:

- SetTID
- GetTID
- GetPLDMVersion
- GetPLDMTypes
- GetPLDMCommands

5.2 PLDM for Platform Monitoring and Control Specification Commands

The BMC implements the following PLDM for Platform Monitoring and Control Specification commands:

- SetTID
- GetTID
- GetTerminusUID
- SetNumericSensorEnable
- GetSensorReading
- GetSensorThresholds
- SetSensorThresholds
- RestoreSensorThresholds
- GetSensorHysteresis
- SetSensorHysteresis
- SetStateSensorEnables
- GetStateSensorReadings
- SetNumericEffecterEnable
- SetNumericEffecterValue
- GetNumericEffecterValue
- SetStateEffecterEnables
- SetStateEffecterStates
- GetStateEffecterStates
- GetPDRRepositoryInfo
- GetPDR

5.3 PLDM for FRU Data Specification Commands

The BMC implements the following PLDM for FRU Data Specification commands:

- GetFRURecordMetadata
- GetFRURecordTable
- GetFRURecordByOption

5.4 PLDM for Vendor Defined Commands

The BMC implements the following PLDM Vendor Defined commands.

5.4.1 Vendor Defined Command Specification Version

The version for the commands described below is 1.2.0 (0xF1F2F000).

Table 3: Vendor defined commands

Command	Command code
GetVersion	0x01
ClockReload	0x05
ClockReadFreq	0x06
ClockReadInfo	0x08
BLCmd	0x0C
UploadFile	0x13
DownloadFile	0x14
FileHash	0x15
DirectoryList	0x16
FpgaLoad	0x17
FileUnlink	0x18
FileMove	0x19
MakeDir	0x1A
LogWrite	0x1C
SetRTC	0x1D

5.4.2 GetVersion

The GetVersion command will return the current BMC Firmware version and other information.

Table 4: GetVersion command

Byte	Type	Request Data
0	uint8	**Device** 0x2 MICROCONTROLLER_VERSION
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, ERROR_INVALID_DATA}
1:4	uint32	**Version**
Variable	string list	**infoStrings** List of NULL terminated ASCII encoded strings

5.4.3 ClockReload

The ClockReload message directs the BMC to reload the specified clock program.

Table 5: ClockReload command

Byte	Type	Request Data
0	uint8	**ClockDevice** DeviceNumber
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, ERROR, ERROR_INVALID_DATA}

5.4.4 ClockReadFreq

The ClockReadFreq message retrieves the frequency info for a specific channel on a clock device. This is the information stored in the program file and not a live value from the clock device.

Table 6: ClockReadFreq command

Byte	Type	Request Data
0	uint8	**ClockDevice** DeviceNumber
1	uint8	**Frequency** Clock channel
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, ERROR}
Variable	-	**FrequencyData** ASCII encoded string

5.4.5 ClockReadInfo

The ClockReadInfo message retrieves clock info related to a clock device. This message needs to be used 1 or more times until the returned count is less than the requested count.

Table 7: ClockReadInfo command

Byte	Type	Request Data
0	uint8	**ClockDevice** DeviceNumber
1:2	uint16	**Count** (max 256)
3:4	uint16	**Offset**
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, ERROR}
1:2	uint16	**Count** Number of bytes in string
Variable	-	**Info** ASCII encoded info string

5.4.6 BLCmd

Commands related to BMC upgrade.

Table 8: BLCmd command

Byte	Type	Request Data
0	uint8	**Command** Bootloader Command
Variable	-	**cmdRequest** Bootloader command request
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {BL_CMD_COMPLETION_CODE}
1	uint8	**Command** Bootloader Command
Variable	-	**cmdResponse** Bootloader command response

5.4.7 Bootloader Command Codes

Table 9: Bootloader Command Codes

Command	Code
Reboot	0x06
GetVersion	0x07
Upgrade	0x08

Table 10: Bootloader Completion Codes

Name	Code	Description

SUCCESS	0x00	The Bootloader command was accepted and completed normally.
ERROR	0x80	Generic failure.
CMD_NOT_FOUND	0x81	Bootloader command not found
NOT_SUPPORTED	0x82	Request is not supported
FLASH_PROTECT	0x83	Flash is protected due to too frequent upgrades.

5.4.8 Reboot Command

Table 11: Reboot command

Byte	Type	Request Data
N/A		
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, ERROR}

5.4.9 GetVersion Command

Table 12: GetVersion command

Byte	Type	Request Data
N/A		
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, ERROR}
1	uint8	**ActiveSlot**
2:9	FirmwareVersion	**Slot1Version**
10:13	uint32	**Slot1Size**
14:21	FirmwareVersion	**Slot2Version**
22:25	uint32	**Slot2Size**

5.4.10 FirmwareVersion Structure

Table 13: FirmwareVersion structure

Byte	Type	Data
0	uint8	**MajorVersion**
1	uint8	**MinorVersion**
2:3	uint16	**Revision**
4:7	uint32	**BuildNumber**

5.4.11 Upgrade Command

Table 14: Upgrade command

Byte	Type	Request Data

0	uint8	**Slot**
1:4	uint32	**DataTransferHandle**
5	enum8	**TransferFlag** {Start=0x00, Middle=0x01, End=0x04, StartAndEnd=0x05}
6:7	uint16	**DataLength**
8:variable	-	**Data**
last	uint8[32]	**SHA256** SHA-256 hash of the transferred data. This field is only present when transferFlag = End or StartAndEnd.
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, NOT_SUPPORTED, FLASH_PROTECT}
1:4	uint32	**NextDataTransferHandle**

5.4.12 UploadFile

Put a file to the BMC Filesystem

Table 15: UploadFile command

Byte	Type	Request Data
0:3	uint32	**FileNameLength**
4:7	uint32	**Offset**
8:11	uint32	**DataLength**
12:-	char	**FileName**
12+FileNameLen:-	uint8	**FileData**
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, ERROR}

5.4.13 DownloadFile

Get a file from the BMC filesystem. Download is complete when ResponseCount is less than the RequestCount.

Table 16: DownloadFile command

Byte	Type	Request Data
0:3	uint32	**offset** Indicates where in the file the data is located.
4:7	uint32	**RequestCount** Maximum number of bytes to return. The responder may return less.
8:11	uint32	**FilePathLength** Length of the FilePath field
12:-	string	**FilePath**
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, FILE_NOT_FOUND}
1:4	uint32	**offset**
5:8	uint32	**ResponseCount** Length of data in the response
9:-	uint8	**Data** File data

5.4.14 FileHash

Calculate and return the SHA-256 hash of a file on the BMC filesystem.

Table 17: FileHash command

Byte	Type	Request Data
0:3	uint32	**FilePathLength** Length of the FilePath field
4:-	string	**FilePath**
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, FILE_NOT_FOUND}
1:33	uint8	**Sha256Hash**

5.4.15 DirectoryList

Get a directory list from the BMC filesystem.

Table 18: DirectoryList command

Byte	Type	Request Data
0:1	uint16	**PathLength** Length of the FilePath field
2:-	string	**DirectoryPath**
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, FILE_NOT_FOUND}
1	uint8	**NumberOfEntries**
2:	DirectoryEntry	**DirectoryEntryList** Array of DirectoryEntry structures

5.4.16 DirectoryEntry

Directory Entry structure

Table 19: DirectoryEntry structure

Byte	Type	Data
0	enum8	**EntryType** {file, directory}
1	uint8	**EntrySize** Lenth of this DirectoryEntry in bytes
2:5	uint32	**FileSize** Size of the File in bytes. If type is Directory this field will be 0.
6	uint8	**NameLength**
7:	string	**Name** File/Directory name

5.4.17 FpgaLoad

Load an FPGA bitstream to flash or to the FPGA directly.

Table 20: FpgaLoad command

Byte	Type	Request Data
0	enum8	**TransferFlag** {Start=0x00, Middle=0x01, End=0x04, StartAndEnd=0x05}
1:4	uint32	**FlashOffset** (`0x46504741` means write to FPGA directly) This value should stay the same for the whole series of write messages from start to end.
5:6	uint16	**DataLength** Number of bytes in this message
7:-	uint8	**Data**
Byte	**Type**	**Response Data**
0	enum8	**completionCode**

5.4.18 FileUnlink

Delete a file in the BMC Filesystem.

Table 21: FileUnlink command

Byte	Type	Request Data
0:1	uint16	**FilePathLength** Length of the FilePath field
2:	string	**FilePath**
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, FILE_NOT_FOUND}

5.4.19 FileMove

Move a file on in the BMC Filesystem.

Table 22: FileMove command

Byte	Type	Request Data
0:1	uint16	**SourceFilePathLength** Length of the Source FilePath field
2:3	uint16	**DestinationFilePathLength** Length of the Destination FilePath field
4:variable	string	**SourceFilePath**
variable:variable	string	**DestinationFilePath**
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, FILE_NOT_FOUND, DESTINATION_INVALID}

5.4.20 MakeDir

Make a directory in the BMC filesystem.

Table 23: MakeDir command

Byte	Type	Request Data
0:3	uint32	**DirNameLength**
4	bool	**MakeParents**
5:-	char	**DirName**
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, ERROR}

5.4.21 LogWrite

Write an entry into the BMC log.

Table 24: LogWrite command

Byte	Type	Request Data
0	enum8	**LogLevel** {INF, ERROR}
1	uint8	**WriteLength**
2:-	string	**LogMessage**
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, ERROR}

5.4.22 SetRTC

Set the BMC RTC (Real Time Clock) time. Time is in timestamp104 format.

Table 25: SetRTC command

Byte	Type	Request Data
0:12	timestamp104	**Time**
Byte	**Type**	**Response Data**
0	enum8	**completionCode** Possible values: {SUCCESS, ERROR}

6 Numeric Sensors and Thresholds

These thresholds are nominal and can be offset due to sensor accuracy and tolerance. Also, these values may be different in different versions of the BMC firmware. Use the **bw_card_monitor** tool to display the current thresholds being used.

Table 26: IA-720i Sensors and Thresholds

Name	Unit	Fatal Low	Critical Low	Warning Low	Warning High	Critical High	Fatal High
FPGA Temperature	C	-	5	6	87.5	100	105
QSFP0 Temperature	C	-	5	6	65	70	-
QSFP1 Temperature	C	-	5	6	65	70	-
Front Temperature	C	2	5	6	75	80	85
Rear Temperature	C	2	5	6	75	80	85
FPGA Fabric Temperature	C	-	5	6	87.5	100	105
FPGA FTile Temperature	C	-	5	6	87.5	100	105
BMC Temperature	C	-	-	-	90	105	-
Total Input Power	W	-	-	-	200	225	-
VCC 12V0 PCIe Voltage	V	9.8	11.04	11.2	12.8	12.96	-
VCC 12V0 PCIe Current	A	-	-	-	5.25	5.5	-
VCC 12V0 PCIe Power	W	-	-	-	63	66	82.5
VCC 3V3 PCIe Voltage	V	-	3	3.15	3.5	3.6	-
VCC 3V3 PCIe Current	A	-	-	-	2.7	3	-
VCC 3V3 PCIe Power	W	-	-	-	8	9	-
VCC 12V0 EXT1 Voltage	V	9.8	11.04	11.2	12.5	12.6	-
VCC 12V0 EXT1 Current	A	-	-	-	11.25	12.5	-
VCC 12V0 EXT1 Power	W	-	-	-	130	150	187.5
VCC Core Voltage	V	-	0.679	0.69	.92	.927	-
VCC Core Current	A	-	-	-	90	95	100
SmartVID Voltage	V	-	.679	.69	.92	.927	-

Table 27: IA-860m Sensors and Thresholds

Name	Unit	Fatal Low	Critical Low	Warning Low	Warning High	Critical High	Fatal High
FPGA Temperature	C	0	5	10	87.5	100	105

QSFP0 Temperature	C	0	5	10	65	67.5	70
QSFP1 Temperature	C	0	5	10	65	67.5	70
QSFP2 Temperature	C	0	5	10	65	67.5	70
Front Temperature	C	0	5	10	75	80	85
Rear Temperature	C	0	5	10	75	80	85
FPGA Fabric Temperature	C	0	5	10	87.5	100	105
FPGA FTile Temperature	C	0	5	10	87.5	100	105
BMC Temperature	C	-	-	-	-	-	-
VCC 12V0 PCIe Voltage	V	9.6	11.04	11.3	12.8	12.96	13.5
VCC 12V0 PCIe Current	A	-	-	-	5.25	5.5	6.05
VCC 12V0 PCIe Power	W	-	-	-	63	66	72.6
VCC 12V0 EXT1 Voltage	V	9.6	11.04	11.2	12.5	12.6	13.5
VCC 12V0 EXT1 Current	A	-	-	-	6	6.25	6.9
VCC 12V0 EXT1 Power	W	-	-	-	70	75	82.5
Total Input Power	W	-	-	-	333	366	402.6
VCC Core Voltage	V	0.65	0.68	0.69	0.91	0.927	1.1
VCC Core Current	A	0	-	-	225	240	264
QSFP Current	A	-	-	-	22	36	50
QSFP Voltage	V	3	3.102	3.168	3.432	3.498	3.63
SmartVID Voltage	V	-	.66	.67	.91	.927	-

Table 28: IA-780i Sensors and Thresholds

Name	Unit	Fatal Low	Critical Low	Warning Low	Warning High	Critical High	Fatal High
FPGA Temperature	C	0	5	10	87.5	100	105
QSFP0 Temperature	C	-5	0	10	65	67.5	70
QSFP1 Temperature	C	-5	0	10	65	67.5	70
Front Temperature	C	5	-	-	75	80	85
Rear Temperature	C	5	-	-	75	80	85
FPGA Fabric Temperature	C	0	-	-	87.5	100	105
FPGA FTile Temperature	C	0	-	-	87.5	100	105
BMC Temperature	C	-	-	-	-	-	-
DIMM0 Temperature	C	-	-	-	85.5	95	105
DIMM1 Temperature	C	-	-	-	85.5	95	105
Total Input Power	W	-	-	-	140	150	165
VCC 12V0 PCIe Voltage	V	9.6	10.2	10.5	12.8	12.96	13.5
VCC 12V0 PCIe Current	A	-	-	-	5.25	5.5	6.05
VCC 12V0 PCIe Power	W	-	-	-	63	66	72.6
VCC 3V3 PCIe Voltage	V	2.64	2	3.15	3.5	3.6	3.96
VCC 3V3 PCIe Current	A	-	-	-	2.7	3	3.3
VCC 3V3 PCIe Power	W	-	-	-	8	9	10
VCC 12V0 EXT1 Voltage	V	9.6	10.2	10.5	12.5	12.6	13.23
VCC 12V0 EXT1 Current	A	-	-	-	6	6.25	6.9
VCC 12V0 EXT1 Power	W	-	-	-	70	75	82.5
VCC Core Voltage	V	0.65	0.66	0.67	.91	.927	1.1
VCC Core Current	A	0	-	-	-	-	120
SmartVID Voltage	V	-	.66	.67	.91	.927	-

Table 29: IA-440i Sensors and Thresholds

Name	Unit	Fatal Low	Critical Low	Warning Low	Warning High	Critical High	Fatal High
FPGA Temperature	C	-	-	-	87.5	90	100
QSFP0 Temperature	C	0	-	-	65	67.5	70
Total Input Power	W	-	-	-	-	-	75
VCC 12V0 PCIe Voltage	V	9.6	10.2	10.5	13.3	13.4	13.5
VCC 12V0 PCIe Current	A	-	-	-	6	6.5	7
VCC 12V0 PCIe Power	W	-	-	-	-	-	66
VCC 3V3 PCIe Voltage	V	2.64	-	-	-	-	3.96
VCC 3V3 PCIe Current	A	-	-	-	-	-	-
VCC 3V3 PCIe Power	W	-	-	-	-	-	9
VCC Core Voltage	V	0.65	0.66	0.67	0.905	0.91	0.927
VCC Core Current	A	0	-	-	61	65	68
SmartVID Voltage	V	-	.66	.67	.905	.91	-
Front Temperature	C	5	-	-	90	95	100

6.1 State Sensors

State can be read with the **bw_card_monitor** tool.

6.1.1 Board Power

Current state of the board power supplies.

State Name	Effect
On	FPGA power supplies turned on.
Off	FPGA power supplies turned off.

6.1.2 FPGA Reset

Current state of the user FPGA reset from the BMC to FPGA.

State Name	Effect
Unasserted	FPGA User reset line is unasserted (high). (default)
Asserted	FPGA User reset line is asserted (low).

6.1.3 MicroController Reset Cause

Reset cause of the last BMC Processor reset.

State Name	Effect
Power on reset	Power on reset
External reset	External reset
Security reset	Internal processor security violation
Watchdog reset	Watchdog timed out
On-chip debug	Debug pod reset
Software reset	Software reset

Temperature reset	Internal processor exceeded maximum threshold
User reset	User reset

6.1.4 QSFPx Present

State of a QSFP Present signal. There will be a separate sensor for each QSFP (i.e QSFP0 Present, QSFP1 Present, etc)

State Name	Effect
Present	QSFP is present.
Removed	QSFP is removed

6.1.5 QSFPx Interrupt

State of a QSFP Interrupt signal. There will be a separate sensor for each QSFP (i.e QSFP0 Interrupt, QSFP1 Interrupt, etc)

State Name	Effect
Asserted	QSFP Interrupt is asserted
Unasserted	QSFP Interrupt is not asserted

6.1.6 IA-440i State Sensors

SiT95148 Loss Of Lock

State of the SiT95148 clock chip Loss of Lock signal.

State Name	Effect
Asserted	LOL is asserted
Unasserted	LOL is not asserted

SiT95148 Interrupt

State of the SiT95148 clock chip Interrupt signal.

State Name	Effect
Asserted	Interrupt is asserted
Unasserted	Interrupt is not asserted

6.1.7 IA-780i State Sensors

SiT95148 Loss Of Lock

State of the SiT95148 clock chip Loss of Lock signal.

State Name	Effect
Asserted	LOL is asserted
Unasserted	LOL is not asserted

SiT95148 Interrupt

State of the SiT95148 clock chip Interrupt signal.

State Name	Effect
Asserted	Interrupt is asserted
Unasserted	Interrupt is not asserted

12v EXT1 Cable

State of the external 12V power cable.

State Name	Effect
75_Watt	Attached cable is a 75 Watt power cable
150_Watt	Attached cable is a 150 Watt power cable
Removed	Power cable is removed

DIMMx Present

DIMM present or removed. There are two DIMM slots so there are two sensors (DIMM0 Present and DIMM1 Present)

State Name	Effect
Present	DIMM present
Removed	DIMM removed

6.1.8 IA-860m State Sensors

SiT95148_(0/1) Loss Of Lock

State of the SiT95148 clock chip Loss of Lock signal.

State Name	Effect
Asserted	LOL is asserted
Unasserted	LOL is not asserted

12v EXT1 Cable

State of the external 12V power cable.

State Name	Effect
150_Watt	Attached cable is a 150 Watt power cable
300_Watt	Attached cable is a 300 Watt power cable
450_Watt	Attached cable is a 450 Watt power cable
600_Watt	Attached cable is a 600 Watt power cable
Removed	Power cable is removed

6.1.9 IA-720i State Sensors

SiT95316 Loss Of Lock

State of the SiT95148 clock chip Loss of Lock signal.

State Name	Effect
Asserted	LOL is asserted
Unasserted	LOL is not asserted

12v EXT1 Cable

State of the external 12V power cable.

State Name	Effect
75_Watt	Attached cable is a 75 Watt power cable
150_Watt	Attached cable is a 150 Watt power cable
Removed	Power cable is removed

6.2 State Effecters

The card state can be controlled via PLDM state effecters using the **bw_bmc_pldm_effecter** command.

6.2.1 Board Power

Controls board supplies that power the FPGA. Can be turned on and off to blank the FPGA.

State Name	Effect
On	FPGA power supplies turned on.
Off	FPGA power supplies turned off.

6.2.2 FPGA Reset

Controls a user FPGA reset from the BMC to FPGA; this is separate from the PCIe reset.

State Name	Effect
Unasserted	FPGA User reset line is unasserted (high) (default)
Asserted	FPGA User reset line is asserted (low).
Pulsed	BMC goes to asserted state, then to unasserted.

6.2.3 PCIe SMBus interface

Controls the PCIe SMBus fingers interface to the card to BMC. The value of this effecter is non-volatile.

State Name	Effect
Enabled	PCIe SMBus BMC interface is enabled.
Disabled	PCIe SMBus BMC interface is disabled.

6.2.4 SmartVID polling

This effecter is only present on Intel FPGA cards.

When the FPGA on the card is configured, the BMC must regularly read the SmartVID voltage from the FPGA to determine the input core voltage requested. This can be disabled before JTAG reprogramming to increase stability with various versions of Quartus Programmer.

State Name	Effect
Enabled	BMC will regularly read FPGA desired input core voltage and adjust to match. (default)
Disabled	BMC will stop reading FPGA desired input core voltage and cease all I^2C traffic to the FPGA SDM.

6.2.5 QSFPx Reset

Assert or Deassert the reset line to a QSFP. There will be a separate effecter for each QSFP (i.e QSFP0 Reset, QSFP1 Reset, etc)

State Name	Effect
Asserted	BMC will assert the QSFP Reset.
Unasserted	BMC will deassert the QSFP Reset. (default)

6.2.6 QSFPx LP

Assert or Deassert the LPMode line to a QSFP. There will be a separate effecter for each QSFP (i.e QSFP0 LP, QSFP1 LP, etc)

State Name	Effect
Asserted	BMC will assert the QSFP LPMode line.
Unasserted	BMC will deassert the QSFP LPMode. (default)

6.2.7 FPGA Telemetry

Enable or Disable the FPGA Telemetry interface.

State Name	Effect
Enable	BMC will enable the FPGA telemetry interface. (default)
Unasserted	BMC will disable the FPGA telemetry interface.

6.2.8 JTAG Mux

Controls how the FPGA JTAG Mux is configured.

State Name	Effect
USB	FPGA JTAG is connected to the FTDI USB. (default)
Breakout	FPGA JTAG is connected to the Breakout interface (ribbon cable/pico lock).
BMC	FPGA JTAG is connected to the BMC.

6.3 Factory Reset

The BMC supports a Factory Reset function that will remove any customer installed data. Data removed when doing a Factory Reset includes:

- All customer FPGA bitstream files
- Any customer custom clock programming files
- Non-volatile ERROR Logs

Instructions for performing a factory reset can be found in the card's Hardware Reference Guide.

7 Status LEDs

The BMC has two status LEDs that show the status of the BMC. There is a red and a green LED. The location of the LEDs is card-specific and can be found in the card's Hardware Reference Guide. The meaning of the LEDs is shown in the following table:

Table 30: LED States

State	Green LED	Red LED
Power Off-Ready	OFF	Blink 2Hz
Power On	Blink 2Hz	OFF
Power Off-Fault	OFF	Blink 2Hz
Power Off-User	ON	Blink 2Hz
Bootloader	Blink 0.5Hz	Opposite of Green LED
Factory Reset	See Description	See Description

7.1 LED State Description

- **Power Off-Ready** – Waiting for input power, will go to Power On when possible
- **Power On** – Card/FPGA power supplies are on
- **Power Off-Fault** – Card/FPGA powered off due to fault. Will go to Power Off-Ready if input powers go off, or user requests power up.
- **Power Off-User** – Card FPGA powered off due to user request. Will stay this way until told to power back up by user request. Then go to Power Off-Ready.
- **Bootloader** – BMC is in bootloader mode, no valid BMC image found, Power is off.
    - See the DFU Loading/Recovery section on how to load a valid BMC image.
- **Factory Reset** – BMC is performing a factory reset. Red LED will blink twice (2Hz), then Green LED twice (2Hz). Repeat

8 FPGA Loading

The BMC will attempt to load the FPGA with a bitstream stored in flash after power is up and stable and the clock(s) have been programed. The BMC supports a fallback mechanism where the user can specify a priority for all or a subset of bitstreams stored in flash. If there are no user bitstreams in flash or none of the bitstreams in flash have a priority the BMC will load the BittWare Card Test FPGA bitstream (if programmed on the card).

8.1 FPGA Bitstream Priority

Bitstream load priority is a numerical value with 0 being the highest priority. Priority value can be any value from 0 - 255. Setting a bitstream priority to -1 will remove the priority on the bitstream and the BMC will ignore that bitstream when selecting an initial or fallback bitstream to load. Duplicate priority on multiple bitstreams is not allowed.

8.2 BittWare FPGA Utility

BittWare provides a utility **bw_bmc_fpga_load.** This utility can be used to upload, remove and set priority on user bitstreams in flash. There are some additional runtime commands shown below. If the user needs to configure FPGA flash outside of using this utility using raw PLDM messages, please contact BittWare support for guidance. Further help on the **bw_bmc_fpga_load** utility can be found in the SDK documentation.

8.3 FPGA Commanded Load

To request the BMC to load a bitstream at runtime that will overwrite the currently loaded bitstream, use the following command:

bw_bmc_fpga_load load <bitstream name>

This command will unload the FPGA (if it is already loaded) and then load the FPGA with the requested bitstream that is stored in flash.

8.4 FPGA Streaming Load

To stream an RBF bitstream to the FPGA directly over USB, use the host utility:

bw_bmc_fpga_load stream <RBF file>

8.5 BittWare Card Test Bitstream

Early Access cards do not have the BittWare Card Test bitstream loaded in flash. If this functionality is desirable on these cards, the user can program the bitstream in the user flash and set the priority to be lower (higher value) than all other user bitstreams.

9 DFU Loading/Recovery

If the BMC image is corrupted or no valid BMC image can be found, the bootloader will fall back to a DFU loading mechanism. When the bootloader is in this state, the card USB Device ID is (0x1117), and the BMC LEDs will be in the “Bootloader” state.

9.1.1 DFU Loading Procedure

First add a DFU suffix to the BMC image file:

```
$ dfu-suffix -v 2525 -p 1117 -a zephyr.signed.bin
Now load the image:
$ dfu-util -D zephyr.signed.bin -a 1
```

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